Linear Algebra with Applications in Automatic Control

School of Automation,
Beijing Institute of Technology

Fall Semester

Outline

- Introduction of this course
- A big picture of linear algebra
- Go into details of Chapter 1
- A few things to discuss with you

Introduction

- Course size: 48 Lecture hours, Week 1-Week 8.

- Lecturer: Dr. Zhuoyue Song

Contact Details: Room 502, Floor 5, No. 6 Building;

E-mail: szhuoyue@bit.edu.cn

- Text: Linear Algebra with Applications in Automatic Control

Author：Prof. Qinghe Wu, National Defense Industry Press

PS: This book can be bought at Room 307, No.6 Building (Teacher Yali Lu)

Introduction

- Exercises, Exams and Grades

  Scores= Exercises* 30% + Exams* 70%

  Exercises: turn in by groups or individuals

- Tips:

1. Many definitions and theorem statements:

Don't recite everything, understand the key idea behind the statements.

2. Sometimes you may find linear algebra is quite abstract,

   remember abstract algebra arises by relaxing the axioms of a vector space, leading to a number of generalizations,

   so try to understand it from the vector-space point of view, and think the easiest situation first, eg. Project into 2 or 3-demensions.

Introduction

Tips-continued

3. This course helps guide you to the door of linear algebra, for deep understanding and extensive reading, please refer to the following list of books/videos:

- MIT-open course on linear algebra

  This covers matrix theory and linear algebra, emphasizing topics useful in other disciplines such as physics, economics and social sciences, natural sciences, and engineering.

- Matrix Analysis by Roger A. Horn

- <系统与控制理论中的线性代数>---黄琳著

  Also a classical book which offers mathematical basics for control theory.

A big picture of Linear Algebra

Z. Song

- Linear algebra is the branch of mathematics concerning vector spaces, often finite or countable infinite dimensional, as well as linear mappings between such spaces. Such an investigation is initially motivated by a system of linear equations in several unknowns. Such equations are naturally represented using the formalism of matrices and vectors.

- Linear algebra is central to both pure and applied mathematics. For instance, abstract algebra arises by relaxing the axioms of a vector space, leading to a number of generalizations.

  Functional analysis studies the infinite-dimensional version of the theory of vector spaces. Combined with calculus, linear algebra facilitates the solution of linear systems of differential equations. Techniques from linear algebra are also used in analytic geometry, engineering, physics, natural sciences, computer science, and the social sciences (particularly in economics). Because linear algebra is such a well-developed theory, nonlinear mathematical models are sometimes approximated by linear ones.

A big picture of this course

- Two parts:

1. Abstract linear algebra (the first four chapters)

collect fundamental material in standard courses on linear algebra

Chapter 1: Linear space, linear mapping and the associated matrix representation
Chapter 2: Polynomial algebra and polynomial matrices
Chapter 3: Linear transformation (LT): canonical form representation, invariant characteristics
Chapter 4: Linear transformation in unitary spaces: normal transformation, unitary transformation, ect.

2. Emphasizing topics useful in systems and control theory
(the last six chapters)

develop materials that is useful in systems and control theory

A big picture of this course

Z. Song

Part 2- Continued

Chapter 5: Decomposition of linear transformations and matrices

Chapter 6: Norms for vectors and matrices (generalizations of length of vector and size of matrices)

Chapter 7: Functions of matrices

Chapter 8: Matrix-valued functions and applications to differential equations

Chapter 9: Generalized inverses of linear transformations and matrices

Chapter 10: Solutions to matrix equations
(linear matrix equations, Algebraic Riccati Equations)

Chapter 1: Linear Space and Mapping

- What includes in this chapter?

Section 1.1 Basic abstract algebra

Section 1.2 Linear space

Section 1.3 Basis of Linear space

Section 1.4 Linear subspace

Section 1.5 Linear transformations

1.1 Concepts of abstract algebra

- Binary algebraic operation

Definition 1.1 Let $\mathcal{U}$, $\mathcal{V}$ and $\mathcal{W}$ be nonempty sets. If for any $\alpha \in \mathcal{U}$ and $\beta \in \mathcal{V}$ there is a correspondence, denoted $\star$, in $\mathcal{W}$, meaning that there is a unique $\gamma \in \mathcal{W}$ such that $\gamma = \alpha \star \beta$, then the correspondence $\star$ is referred to as a binary algebraic operation from $\mathcal{U} \times \mathcal{V}$ to $\mathcal{W}$. Furthermore, if $\mathcal{U} = \mathcal{V} = \mathcal{W}$, we say that $\star$ is closed on $\mathcal{U}$.

1.1 Concepts of abstract algebra

- An example of a binary algebraic operation

Example 1.1 Let $\mathcal{U} = \mathcal{C}^{n\times m}$, $\mathcal{V} = \mathcal{C}^{m\times p}$, $\mathcal{W} = \mathcal{C}^{n\times p}$. Then the matrix multiplication $\cdot$ is a binary operation from $\mathcal{U} \times \mathcal{V}$ to $\mathcal{W}$.

Note that the elements in

$$\mathcal{U} \times \mathcal{V} = \{ (\alpha , \beta) \ : \ \alpha \in \mathcal{U} , \quad \beta \in \mathcal{V} \}$$

are ordered pairs of the form (α , β) with $\alpha \in \mathcal{U}$ and $\beta \in \mathcal{V}$. From example 1.1 it should be clear that the ordering of (α , β) is very critical, since for this example $\alpha \cdot \beta$ is a performable operation but $\beta \cdot \alpha$ does not make sense if $p \neq n$.

1.1 Concepts of abstract algebra

- Some operation properties

Definitin 1.2 Let the operation $\star$ be closed on $\mathcal{U}$. If for any α, $\beta \in \mathcal{U}$ the following property

$$\alpha \star \beta = \beta \star \alpha$$

is satisfied, we say that $\star$ is commutative on $\mathcal{U}$. If for any α, β, $\gamma \in \mathcal{U}$ the following property

$$\alpha \star (\beta \star \gamma) = (\alpha \star \beta) \star \gamma$$

is satisfied, we say that $\star$ is associative on $\mathcal{U}$.

1.1 Concepts of abstract algebra

- Definition of an algebraic system

Definitin 1.3 An algebraic system is a set $\mathcal{U}$ together with some binary operations that are closed on $\mathcal{U}$ and satisfy some properties.

- Next, we shall introduce three elementary algebraic systems, they are groups, rings and fields

1.1 Concepts of abstract algebra

•Groups

Definition 1.4 A group $\mathcal{G}$ is a nonempty set together with a binary operation $\star$ that is closed on $\mathcal{G}$ and satisfies the following three properties:

1. $\star$ is associative, i.e. $\alpha \star (\beta \star \gamma) = (\alpha \star \beta) \star \gamma$ $\forall \alpha, \beta, \gamma \in \mathcal{G}$;
2. There is an element ε in $\mathcal{G}$ such that

$$\alpha \star \varepsilon = \varepsilon \star \alpha = \alpha \quad \forall \alpha \in \mathcal{G}$$

i.e. there is a unit elment or a zero element;

3. For any $\alpha \in \mathcal{G}$ there exists a $\beta \in \mathcal{G}$ such that $\alpha \star \beta = \beta \star \alpha = \varepsilon$, i.e. any element in $\mathcal{G}$ has its inverse or negative element in $\mathcal{G}$.

1.1 Concepts of abstract algebra

- Abelian group and additive group

If $\star$ is commutative on $\mathcal{G}$, i.e. $\alpha \star \beta = \beta \star \alpha$ $\forall \alpha, \beta \in \mathcal{G}$, then $\mathcal{G}$ is referred to as an Abelian group. Especially, if $\star$ represents addition, then $\mathcal{G}$ is an additive group.

- An theorem

Theorem 1.1 The identity or the zero element in $\mathcal{G}$ is unique; the inverse of any invertible element in $\mathcal{G}$ is unique.

1.1 Concepts of abstract algebra

- Example: additive polynomial group

  - Recall: definition of group—set (polynomials)

    operation (addition)

First, we define polynomials,

then define the addition operation,

finally check if this operation satisfies the three properties.

1.1 Concepts of abstract algebra

- Definition of polynomials

Definition 1.5 Let $\mathcal{F}$ be $\mathcal{R}$ or $\mathcal{C}$, λ be an indeterminate, $n \in \mathcal{Z}_{0,+}$ and $a_i \in \mathcal{F}$, $i = 1, 2, \ldots, n$. The expression $a(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$, $a_i \in \mathcal{F}$, $i = 1, 2, \ldots, n$ is referred to as a formal polynomial with coefficients in $\mathcal{F}$, or a polynomial over $\mathcal{F}$. The set of all polynomials over $\mathcal{F}$ is denoted $\mathcal{F}[\lambda]$.

Notation Denote by $\mathcal{R}$ the set of all real numbers, by $\mathcal{C}$ the set of all complex numbers, by $\mathcal{Z}$ the set of all integers, and by $\mathcal{Z}_{0,+}$ the set of all nonnegative integers respectively. $\mathcal{Z}_+$ is the set of all natural numbers.

1.1 Concepts of abstract algebra

- Definition of equality of polynomials

Definition 1.6 The two polynomials

$$\begin{aligned} a(\lambda) &= a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0\,, \\ b(\lambda) &= b_m\lambda^m + b_{m-1}\lambda^{m-1} + \cdots + b_1\lambda + b_0 \end{aligned}$$

in $\mathcal{F}[\lambda]$ are said to be equal if 1. $m = n$, 2. $a_i = b_i$, $i = 1, 2, \ldots, n$ and denoted $a(\lambda) = b(\lambda)$. If all the coefficients of a polynomial are zero, then we call it a zero polynomial. Zero polynomial is denoted 0.

1.1 Concepts of abstract algebra

- Definition of degree of polynomials

If $a(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$ with $a_n \neq 0$, then we say that $a(\lambda)$ is a polynomial of degree n, n is called the degree of $a(\lambda)$, denoted $\deg a(\lambda) = n$. $a_n\lambda^n$ with $a_n \neq 0$ is called the leading term of $a(\lambda)$ and a_n is called the leading coefficient. A polynomial of degree n with $a_n = 1$ is said to be monic.

1.1 Concepts of abstract algebra

Z. Song

- Definition of addition operation

The sum of $a(\lambda)$ and $b(\lambda)$ is denoted $(a+b)(\lambda)$, and is defined as

$$\begin{aligned} c(\lambda) &= (a+b)(\lambda) = a(\lambda) + b(\lambda) \\ &= (a_n + b_n)\lambda^n + (a_{n-1} + b_{n-1})\lambda^{n-1} + \cdots + (a_1 + b_1)\lambda + (a_0 + b_0) \\ &= c_n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0; \end{aligned}$$

- What shall we do if the degrees are not equal?

1.1 Concepts of abstract algebra

- Definition of addition operation-continued

Let

$$\begin{aligned} a(\lambda) &= a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0\,, \\ b(\lambda) &= b_m\lambda^m + b_{m-1}\lambda^{m-1} + \cdots + b_1\lambda + b_0 \end{aligned}$$

be both in $\mathcal{F}[\lambda]$. Without loss of generality, we assume that $n \geq m$. Then by inserting zero terms, $b(\lambda)$ can be represented as

$$b(\lambda) = b_n\lambda^n + \cdots + b_{m+1}\lambda^{m+1} + b_m\lambda^m + b_{m-1}\lambda^{m-1} + \cdots + b_1\lambda + b_0$$

with

$$b_n = b_{n-1} = \cdots = b_{m+1} = 0$$

1.1 Concepts of abstract algebra

•Check if the addition operation satisfies the three properties?

- 1. associative
- 2 exists zero element 0
- 3 exists negative element

the negative element $-a_n\lambda^n - a_{n-1}\lambda^{n-1} - \cdots - a_1\lambda - a_0$ for $a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$.

Conclusions:

$\mathcal{F}[\lambda]$ with the addition operation '+' is a group.

1.1 Concepts of abstract algebra

- One example of a group with finite number of elements

  - A four-stage shift register example, see the textbook on Page 3-4, please read yourself, basically, check if the system satisfy the three properties of groups via Definition 1.4

1.1 Concepts of abstract algebra

- Definition of rings

Definition 1.8 A ring $\mathcal{R}$ is a nonempty set together with two binary operations $+$ and $\cdot$, generally referred to as addition and multiplication respectively, that are closed on $\mathcal{R}$ and satisfy the following properties:
1. $\mathcal{R}$ is an additive group; 2. $\cdot$ is associative, meaning that $\alpha\cdot(\beta\cdot\gamma)=(\alpha\cdot\beta)\cdot\gamma\ \forall\alpha,\ \beta,\ \gamma\in\mathcal{R}$;
3. multiplication is left and right distributive over addition, that is, for all $\alpha,\ \beta,\ \gamma\in\mathcal{R}$ there holds

$$\alpha\cdot(\beta+\gamma)=\alpha\cdot\beta+\alpha\cdot\gamma\,,\quad(\beta+\gamma)\cdot\alpha=\beta\cdot\alpha+\gamma\cdot\alpha\,.$$

Furthermore, if multiplication is commutative in $\mathcal{R}$, i.e. $\alpha \cdot \beta = \beta \cdot \alpha \ \forall \alpha, \ \beta \in \mathcal{R}$, then $\mathcal{R}$ is referred to as a commutative ring. A nonzero element $\alpha \in \mathcal{R}$ is referred to as a left (right) zero divisor if $\alpha \cdot \beta = 0$ $(\beta \cdot \alpha = 0)$ for some nonzero $\beta \in \mathcal{R}$, or a left (right) annihilating element. If α is both a left and right zero divisor, then it is called a zero divisor. A commutative ring $\mathcal{R}$ is then referred to as an integral domain if it contains no zero divisors. If there is an element ε_L (ε_R) in $\mathcal{R}$ such that $\varepsilon_L \cdot \alpha = \alpha$ $(\alpha \cdot \varepsilon_R = \alpha)$ $\forall \alpha \in \mathcal{R}$, then ε_L (ε_R) is referred to as a left (right) unit for $\mathcal{R}$. If ε is both a left and right unit for $\mathcal{R}$, then it is called a unit for $\mathcal{R}$. For $\alpha \in \mathcal{R}$, if there exists $\alpha^{-1} \in \mathcal{R}$ such that $\alpha \cdot \alpha^{-1} = \alpha^{-1} \cdot \alpha = \varepsilon$, then α^{-1} is referred to as the inverse of α.

1.1 Concepts of abstract algebra

- An theorem

Theorem 1.2 If there is a unit in $\mathcal{R}$, then the unit is unique. If $\alpha \in \mathcal{R}$ has its inverse α^{-1} such that $\alpha \cdot \alpha^{-1}$ is the unit, then the inverse α^{-1} is unique.

- Proof is simple, basically suppose there exist two units and two inverses, and prove they are equally respectively.

1.1 Concepts of abstract algebra

- An example: polynomial ring

  - Previously, we have given an additive polynomial group, i.e. a polynomial set with the addition operation, then define multiplication of polynomials which satisfies the properties of rings., please read yourself (page 6).

1.1 Concepts of abstract algebra

- Definition of fields

Defintion 1.10 A field is a commutative ring having unit and nonzero elements, with every nonzero element being invertible. A field is denoted by $\mathcal{F}$.

Example The most commonly used fields are the fields of all rational numbers, $\mathcal{R}$, $\mathcal{C}$ and the field of all rational functions.

1.2 Linear spaces

- Definition of linear space over a field

Defintion 1.11 Let $\mathcal{V}$ be a nonempty set, $\mathcal{F}$ be a field. A binary operation "$+$", called the addition, is closed on $\mathcal{V}$, that is, for any ordered pair $(\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathcal{V} \times \mathcal{V}$, there exists a unique element $\boldsymbol{\nu} \in \mathcal{V}$, called the sum of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, such that $\boldsymbol{\nu} = \boldsymbol{\alpha} + \boldsymbol{\beta}$. $\mathcal{V}$, together with the addition, forms an additive group. Furthermore, a binary operation, called the scalar multiplication and defined on $\mathcal{F} \times \mathcal{V}$, is closed on $\mathcal{V}$, that is, for any $(k, \boldsymbol{\alpha}) \in \mathcal{F} \times \mathcal{V}$, there exists a unique element $\boldsymbol{\eta} \in \mathcal{V}$, called the scalar multiplication of k and $\boldsymbol{\alpha}$, such that $\boldsymbol{\eta} = k\boldsymbol{\alpha}$. Scalar multiplication satisfies the following four axioms:

1.2 Linear spaces

- Definition of linear space over a field-continued

1. $1 \cdot \boldsymbol{\alpha} = \boldsymbol{\alpha}$,
2. $k(l\boldsymbol{\alpha}) = (kl)\boldsymbol{\alpha}$,
3. $(k+l)\boldsymbol{\alpha} = k\boldsymbol{\alpha} + l\boldsymbol{\alpha}$,
4. $k(\boldsymbol{\alpha}+\boldsymbol{\beta}) = k\boldsymbol{\alpha} + k\boldsymbol{\beta}$, for all l, $k \in \mathcal{F}$ and $\boldsymbol{\alpha}$, $\boldsymbol{\beta} \in \mathcal{V}$. Such a nonempty set $\mathcal{V}$ is called a linear space over the field $\mathcal{F}$.

- Example 1.3:

$\mathcal{C}^{n\times m}$, the set of all $n \times m$ complex matrices, forms a linear space over $\mathcal{C}$, the field of all complex numbers, under the usual matrix summation

$$A+B = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix}$$

$$\triangleq \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \dots & a_{1m}+b_{1m} \\ a_{21}+b_{21} & a_{22}+b_{22} & \dots & a_{2m}+b_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1}+b_{n1} & a_{n2}+b_{n2} & \dots & a_{nm}+b_{nm} \end{bmatrix}$$

- Example 1.3-continued

and the scalar multiplication of $A \in \mathcal{C}^{n\times m}$ and $\alpha \in \mathcal{C}$

$$\alpha \cdot A \triangleq \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \dots & \alpha a_{1m} \\ \alpha a_{21} & \alpha a_{22} & \dots & \alpha a_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ \alpha a_{n1} & \alpha a_{n2} & \dots & \alpha a_{nm} \end{bmatrix}.$$

As a special case of $A \in \mathcal{C}^{n\times m}$ with $m = 1$, the set of all n-dimensional complex vectors

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$$

forms a linear space over $\mathcal{C}$

1.2 Linear spaces

- Example: A linear space of polynomials

  - The set of polynomials with addition operation is an additive group, now define a scalar multiplication as follows:

Definition 1.12 (Scalar multiplication with a number $\alpha \in \mathcal{F}$) For

$$a(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

the scalar multiplication with a number $\alpha \in \mathcal{F}$ is denoted $(\alpha a)(\lambda)$ and referred to as the polynomial

$$(\alpha a)(\lambda) = \alpha a_n\lambda^n + \alpha a_{n-1}\lambda^{n-1} + \cdots + \alpha a_1\lambda + \alpha a_0$$

- Example: A linear space of polynomials-continued

The set of all polynomials

$$a(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + a_{n-1}\lambda^{n-1}$$

with degree not exceeding $n-1$ and coefficients in $\mathcal{F}$, denoted by$\mathcal{F}_{n-1}[\lambda]$, forms a linear space.

It should be noted that the set of all polynomials with degree of $n-1$ is NOT a linear space.

1.2 Linear spaces

- Linear combination

Defintion 1.13 Let $\mathcal{V}$ be a linear space over $\mathcal{F}$, $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_r$ $(r \geq 1)$ be vectors in $\mathcal{V}$, k_1, k_2, ..., k_r be numbers in $\mathcal{F}$. If $\boldsymbol{\alpha}_{r+1} \in \mathcal{V}$ can be represented as

$$\boldsymbol{\alpha}_{r+1} = k_1\boldsymbol{\alpha}_1 + k_2\boldsymbol{\alpha}_2 + \cdots + k_r\boldsymbol{\alpha}_r$$

then we say that $\boldsymbol{\alpha}_{r+1}$ is linearly representable by $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_r$, or $\boldsymbol{\alpha}_{r+1}$ is a linear combination of $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_r$, k_1, k_2, ..., k_r are called the coefficients of the linear combination.

1.2 Linear spaces

- Linear (in)dependency

Definition 1.14 Given the vectors $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ in $\mathcal{V}$ with $s \geq 1$. It there exist $k_1, k_2, \ldots, k_s$ in $\mathcal{F}$ with at least one $k_i \neq 0$, $1 \leq i \leq s$, such that

$$k_1\boldsymbol{\alpha}_1 + k_2\boldsymbol{\alpha}_2 + \cdots + k_s\boldsymbol{\alpha}_s = \mathbf{0} \qquad (1)$$

then $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ are said to be linearly dependent. In other words, if (1) holds if and only if $k_1 = k_2 = \cdots = k_s = 0$, then $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ are linearly independent.

1.2 Linear spaces

- An theorem

Theorem 1.3 If $\alpha_1, \alpha_2, \ldots, \alpha_r$ in $\mathcal{V}$ are linearly independent, while $\alpha_1, \alpha_2, \ldots, \alpha_r, \alpha_{r+1}$ are linearly dependent, then α_{r+1} can be uniquely represented by a linear combination of $\alpha_1, \alpha_2, \ldots, \alpha_r$.

1.3 Basis of a linear space

•1.3.1 The notion of a basis

Definition 1.15 A finite set of vectors $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ in a linear space $\mathcal{V}$ over the field $\mathcal{F}$ is referred to as a basis of $\mathcal{V}$ if they are linearly independent and every vector $\boldsymbol{\alpha}$ in $\mathcal{V}$ is a linear combination of its elements, that is

$$\boldsymbol{\alpha} = k_{\alpha,1}\boldsymbol{\alpha}_1 + k_{\alpha,2}\boldsymbol{\alpha}_2 + \cdots + k_{\alpha,n}\boldsymbol{\alpha}_n$$

$$= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n] \begin{bmatrix} k_{\alpha,1} \\ k_{\alpha,2} \\ \vdots \\ k_{\alpha,n} \end{bmatrix}$$

for all $\boldsymbol{\alpha} \in \mathcal{V}$. The n-dimensional column vector $\boldsymbol{k}_{\boldsymbol{\alpha}} = [k_{\alpha,1} k_{\alpha,2} \ldots k_{\alpha,n}]^T$ is called the representation of $\boldsymbol{\alpha}$ with respect to the basis $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$. $\mathcal{V}$ is called an n-dimensional linear space, denoted $\dim \mathcal{V} = n$.

Denote by $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]$ the basis of a linear space of dimension n.

1.3 Basis of a linear space

- Some explanations

A basis is also called a maximal linearly independent set of the space $\mathcal{V}$.

The $\dim \mathcal{V}$ is the number of vectors in any maximal linearly independent set of $\mathcal{V}$. In this sense, we say also that the rank of an n-dimensional space is n, denoted $\text{rank}\,\mathcal{V} = n$

1.3 Basis of a linear space

- *An example*

Exp: $\mathcal{R}_{n-1}[\lambda]$ is an n-dimensional linear space.

Proof: Via the Exp 1.4 on Pg 8, the set of all polynomials with degree not exceeding $n-1$ and coefficiencts in $\mathcal{R}$ is a linear space. It is clear that the vectors 1, λ, λ^2, ..., λ^{n-1} are all in $\mathcal{R}_{n-1}[\lambda]$ and every $f(\lambda) \in \mathcal{R}_{n-1}[\lambda]$ can be represented as $f(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1}$, i.e. $f(\lambda)$ is a linear combination of these vectors. It suffices then to show that the vectors 1, λ, λ^2, ..., λ^{n-1} in $\mathcal{R}_{n-1}[\lambda]$ are linearly independent.

1.3 Basis of a linear space

- Example-continued

Suppose that there exist k_1, k_2, ..., k_{n-1}, all in $\mathcal{R}$ such that

$$k_0 \cdot 1 + k_1\lambda + \cdots k_{n-1}\lambda^{n-1} = 0\,.$$

Comparing the coefficients of the both sides we conclude that $k_i = 0$ for all $i = 0, 1, 2, \ldots, n$, which implies that 1, λ, λ^2, ..., λ^{n-1} are linearly independent. Thus $\dim \mathcal{R}_{n-1}[\lambda] = n$.

1.3 Basis of a linear space

- Example-continued

It can be readily verified that 1, $\lambda - a$, $(\lambda - a)^2$, ..., $(\lambda - a)^{n-1}$ can be a basis of $\mathcal{R}_{n-1}[\lambda]$ as well, since every polynomial $f(\lambda)$ with degree no more than $n-1$ can be expanded to the Taylor series

$$f(\lambda) = f(a) + f'(a)(\lambda - a) + \frac{f''(a)}{2!}(\lambda - a)^2 + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(\lambda - a)^{n-1}$$

at $\lambda = a$. Thus the representation of $f(\lambda)$ with respect to this basis is

$$f(a), \quad f'(a), \quad \frac{f''(a)}{2!}, \quad \ldots, \quad \frac{f^{(n-1)}(a)}{(n-1)!}.$$

1.3 Basis of a linear space

- 1.3.2 Change of Basis and Transition Matrices
- Basis is not unique, what is relation of different bases and the associated representation of a vector?

Let $[\boldsymbol{\alpha}_1\ \boldsymbol{\alpha}_2\ \ldots\ \boldsymbol{\alpha}_n]$ and $[\boldsymbol{\beta}_1\ \boldsymbol{\beta}_2\ \ldots\ \boldsymbol{\beta}_n]$ be two different bases of $\mathcal{V}$. Since $[\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n]$ form a basis of $\mathcal{V}$, $\boldsymbol{\beta}_i$ can be represented as

$$\boldsymbol{\beta}_i = p_{1i}\boldsymbol{\alpha}_1 + p_{2i}\boldsymbol{\alpha}_2 + \cdots + p_{ni}\boldsymbol{\alpha}_n = [\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n] \begin{bmatrix} p_{1i} \\ p_{2i} \\ \vdots \\ p_{ni} \end{bmatrix},$$

where $i = 1, 2, \ldots, n$. The n equations can be represented by a matrix equation as follows

$$[\boldsymbol{\beta}_1\ \ \boldsymbol{\beta}_2\ \ \ldots\ \ \boldsymbol{\beta}_n] = [\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}}_{\triangleq P}$$

The $n \times n$ matrix P is called the transition matrix from the basis $[\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n]$ to the basis $[\boldsymbol{\beta}_1\ \ \boldsymbol{\beta}_2\ \ \ldots\ \ \boldsymbol{\beta}_n]$.

1.3 Basis of a linear space

- An theorem related to the transition matrix

Theorem 1.4 A transition matrix is invertible.

- The relationship between representation of a vector with respect to different bases?

Let ξ be any vector in $\mathcal{V}$, having the representations $\boldsymbol{x} = [x_1 \quad x_2 \quad \dots \quad x_n]^{\mathsf{T}}$ and $\boldsymbol{y} = [y_1 \quad y_2 \quad \dots \quad y_n]^{\mathsf{T}}$ with respect to the bases $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ and $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, ..., $\boldsymbol{\beta}_n$ respectively, that is

$$\begin{aligned} \boldsymbol{\xi} &= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\,\boldsymbol{x}, \\ \boldsymbol{\xi} &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_n]\,\boldsymbol{y}, \end{aligned}$$

then

$$\boldsymbol{y} = P^{-1}\boldsymbol{x}$$

Summary

- Algebraic systems:

  groups,rings and fields

- Linear space:

  What is it?

  When the elements in a linear space is independent?

- Basis

  What is it? Maximal linearly independent set

  Dimension of linear space? Number of vectors in the basis

  Change of basis, transition matrices, what is relation of different bases and the associated representation of a vector?

Homework

Example 1.8

Example 1.9

Example 1.12